USBSID-Configtool manual

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Disclaimer

I do this stuff in my free time for my enjoyment. Since I like to share my joy in creating this with everyone I try my best to provide a working PCB and Firmware. I am in no way an electronics engineer and can give ***no guarantees*** that this stuff does not break or damage your hardware, computer, phone, or whatever you try to hook it up to. Be sure to take great care when inserting any real MOS SID chips into the board. While everything has been tested with real chips, this is in no way a guarantee that nothing could go wrong. Use of this board and firmware at your own risk! I am in no way responsible for your damaged hardware. That being said, have fun!

About

What is there to say, you can configure your board with this tool and it is self explanatory.
Prefer the command line? A CLI config tool (`cfg_usbsid`) is also available ~ see the `USBSID-Pico-firmware-manual` file for its usage.

Drivers

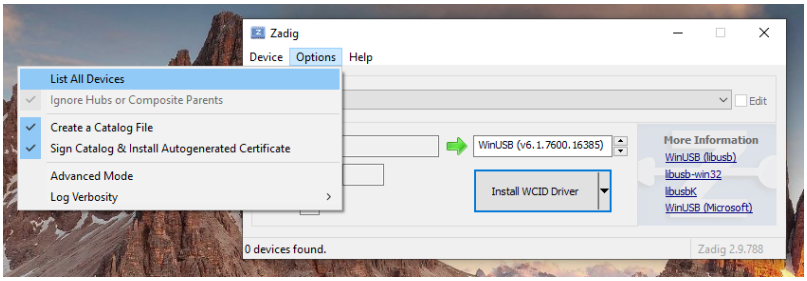
Linux Udev rules

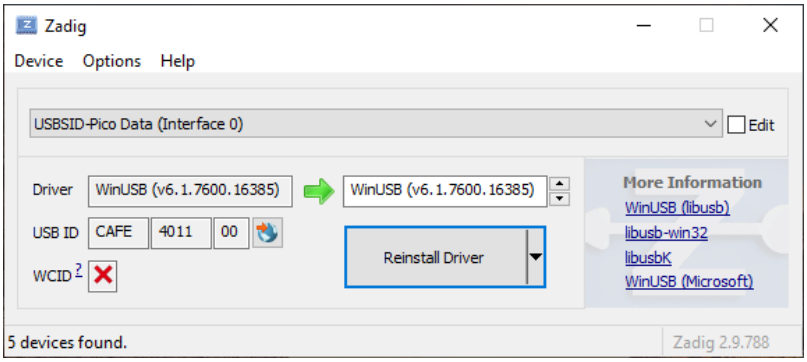
In the examples/udev directory you can find the udev rules that are used on Linux.
This purely an example file that you can use and change to your own needs. Steps required for this to work

```
# Check if you are in the plugdev group with the groups command
groups
# If not, add yourself to the plugdev group, then log out and in
sudo usermod -aG plugdev $USER
# Copy the udev ules file to the correct directory
sudo cp 69-usbsid-permissions.rules /etc/udev/rules.d
# Now reload the udev rules
udevadm control --reload-rules && udevadm trigger
# Not working? Try reloading the service
sudo systemctl restart udev
```

Windows driver

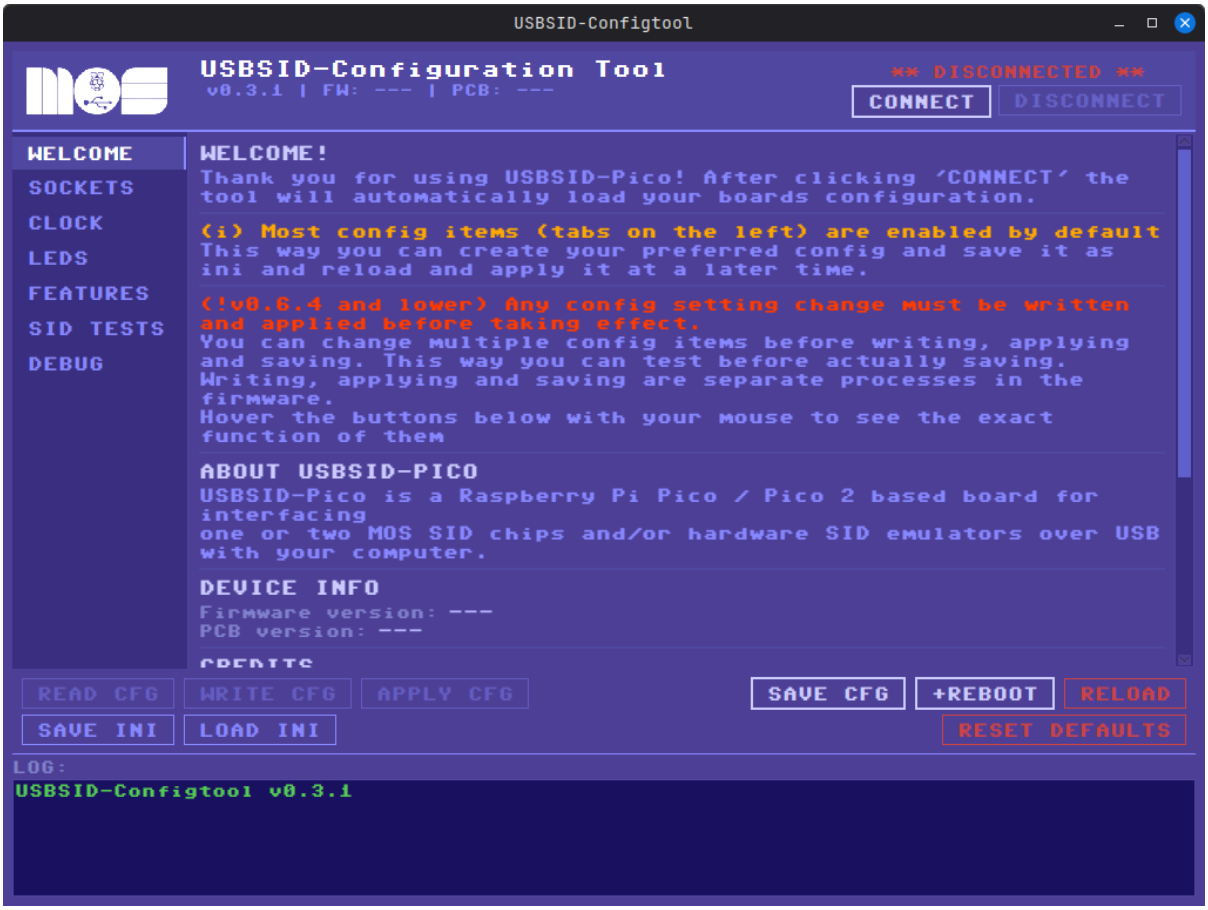
Use Zadig to install the correct driver. Replace the driver for `USBSID-Pico Data` with WinUSB.



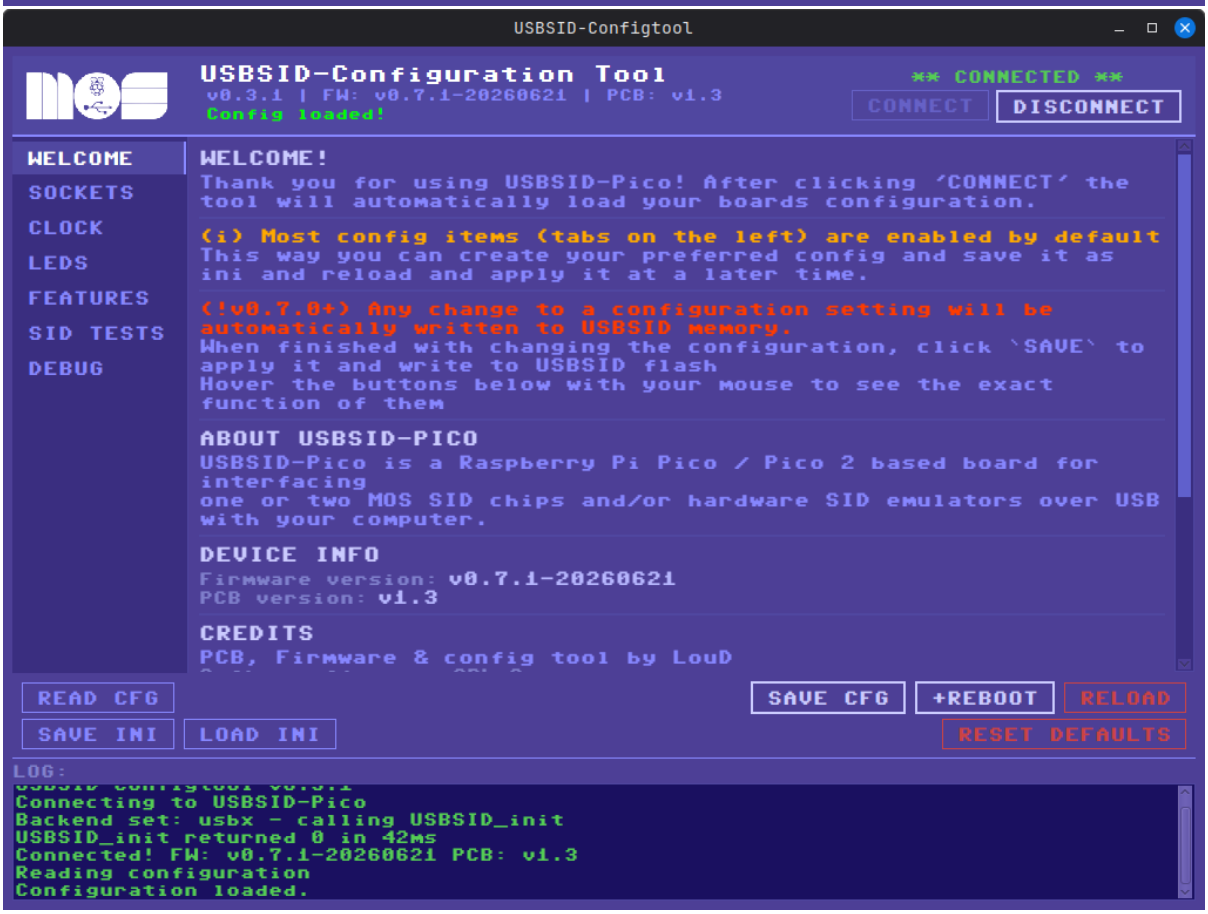


Screens

Welcome screen







v1.5 Sockets confirmation message



Sockets screen





Clock screen



External Clock: display only. The firmware detects an external clock source at startup and reports

it in the configuration, it cannot be set from the Configtool.

Clock link: enables or disables the board clock link (LINK_CLOCK command 0x84, state read from configuration byte 6 bit 1). The firmware saves the configuration and reboots the board when the value changes. The Configtool disconnects and the board must be reconnected afterwards.

LEDS screen



Features screen





USB RX ringbuffer: buffers up to 8 incoming USB packets and prevents cracks on write-heavy tunes. Default on. Listed under ADVANCED next to Preset auto detect, written with the other settings (SET_CONFIG item 17) and stored in the ini file as `rx_ring` in the `[Features]` section. The setting applies after saving the configuration and rebooting the board. Available on firmware

newer than v0.8.1.

Preset auto detect: toggle under ADVANCED, written as SET_CONFIG item 16 (BOARD_PADETECT).

FMOpl SID No.: selector in the FM Opl block on the Sockets screen, enabled when FMOpl is on. The selected SID number (1 to 4) is sent with SET_CONFIG item 9 (BOARD_FMOPL) and must be a SID that has an address assigned. Selecting Unknown (0) while FMOpl is enabled clears the assignment.

SID tests screen



Debug screen



The `RESET SID REGISTERS` button resets all SID registers (same as `cfg_usbsid --reset-sid-registers`).

Development

More information about USBSID-Configtool is available on the github repo

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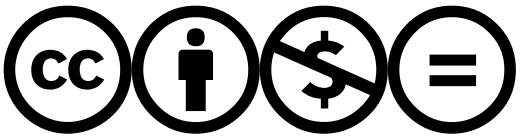
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